LESSON 9.6
REWRITING EQUIVALENT NUMERICAL EXPRESSIONS

LESSON INTRODUCTION

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

LEARNING TARGETS

- I can determine if two numerical expressions are equivalent by rewriting each in the same form.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

ESSENTIAL QUESTIONS

- How can the associative property be used to determine if two expressions are equivalent?
- How can the commutative property be used to determine if two expressions are equivalent?
- How can the distributive property be used to determine if two expressions are equivalent?

LESSON NARRATIVE

In this lesson, students use the associative, commutative, and distributive properties to rewrite numeric expressions. Initially, students think about different ways the associative property of addition can be used. Students learn how to use the commutative, associative, and distributive properties to rewrite numeric expressions involving integers and rational numbers. This review of the properties is then used to help students realize the properties can be used to determine if expressions are equivalent without needing to fully evaluate the expressions. This connection leads directly to determining if two algebraic expressions are equivalent in future lessons.

COMPONENT SUMMARY

9.6.1 WARM-UP *(~5 MIN.)*

Two Truths and a Lie

9.6.2 GUIDED INSTRUCTION *(20 MIN.)*

Explore the properties in the context of equivalent expressions

9.6.3 TRY IT *(15 MIN.)*

Determine if expressions are equivalent without evaluating while naming properties

9.6.4 YOUR TURN! *(5 MIN.)*

Rewrite a given expression two ways using the associative, commutative, and distributive properties

9.6.5 WRAP-UP *(~5 MIN.)*

Determine whether a student correctly used the associative, commutative, or distributive properties to rewrite a numeric expression

RESOURCES

GLOSSARY

Key Terms:

- Commutative property
- Associate property
- Distributive property

Additional Terms: N/A

MATERIALS

- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- (Optional) Materials for graphic organizer for the properties.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may interpret any set of parentheses as the distributive property, even in examples like 3+4+5.
- Students may believe only one property can apply to any set of expressions (question 2 in 9.6.3 Try It).

THINGS TO CONSIDER

- Students have been working with these properties of operations since elementary school, so the 9.6.2 Guided Instruction should not be approached as though this is brand-new content.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

Students continually determine equivalence without evaluating throughout this lesson. They must attend to precision in identifying unique similarities and differences between expressions. Utilize the framework of Notice and Wonder throughout the lesson to create intentional space for students to notice properties and wonder about their equivalence.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Flexibility and Fluency

Fluency takes different forms for different students in a variety of contexts. In this lesson, fluency incorporates flexibility in determining whether expressions are equivalent. Students are intentionally not evaluating, opening up entry points to learning that avoid obstacles of calculating and embrace flexibility in observing, reasoning, and creating multiple pathways to the same answer.

DIFFERENTIATE INSTRUCTION

Students have engaged with commutative, associative, and distributive properties in 5th and 6th grade. Instead of reteaching this content, consider having students produce their working definitions and examples of each property using foldables or other graphic organizers. For example, students may struggle to articulate the difference between what they describe as parentheses "moving around" (associative) and parentheses "dropping" (distributive). Specifically highlight examples and capture student responses that target these misconceptions rather than reteaching the definitions of the properties. Including nonexamples like the "lie" in the 9.6.1 Warm-Up can also be an effective strategy for illustrating the differences between properties.

9.6.1 WARM-UP: TWO TRUTHS AND A LIE

QUESTIONING

If your students enjoyed this activity, consider converting question 7 in 9.6.2 Guided Instruction into Two Truths and a Lie as well, with students creating two equivalent expressions and one "lie" that is not equivalent before trading with a partner to detect the lie.

9.6.1 Warm-Up: Two Truths and a Lie

1. Students were asked to write two truths and a lie, or two true statements and one false statement, about the associative property of addition. One student's work is shown.

Associative Property of Addition

A. $(5+4)+3 \quad 5+(4+3)$

B. $3+(4+5) = (3+4)+5$

C. $(4+3)+5 = 4+(3+5)$

Which statement is the lie? Justify your response.

A is a lie, because it is missing an equal sign between the expression on the left and the expression on the right. It should read: $(5 + 4) + 3 = 5 + (4 + 3)$

9.6.2 Guided Instruction: Rewriting Equivalent Expressions

1. Consider the expressions $18 + 12(30 + 7)$ and $18 + 12(7 + 30)$.

   a. Describe what is different about the expressions.
   The difference is that in the parentheses one expression has $(30 + 7)$ and the other expression has $(7 + 30)$.

   b. Evaluate $18 + 12(30 + 7)$ and $18 + 12(7 + 30)$.

$18 + 12(30 + 7)$ $= 18 + 12(37)$ $= 18 + 444$ $= 462$	$18 + 12(7 + 30)$ $= 18 + 12(37)$ $= 18 + 444$ $= 462$

   c. Are the two expressions equivalent?
   Yes, the two expressions are equivalent.

   d. Was it necessary to evaluate the two expressions to determine whether they were equivalent or not?
   No, because the expression in the parentheses were equivalent and the rest of the expressions are the same.

2. The [○ associative / ○ **commutative** / ○ distributive] property of addition states that the order in which two numbers are added does not matter.

3. For what other mathematical operations does this property hold true?
   The commutative property also holds true for multiplication.

4. Eric thinks the expressions shown are equivalent to each other.
   $18 + 12(30 + 7)$ $\quad 18 + 360 + 84$ $\quad 30(30 + 7)$

   Without evaluating each expression, explain if he is correct.
   The first two expressions are equivalent, because of the distributive property. The third expression is not equal to the first two expressions, because the order of operations tells us we need to multiply before adding.

5. The [○ associative / ○ commutative / ○ **distributive**] property of operations can be used to show that $18 + 12(30 + 7) = 18 + 360 + 84$.

Equivalent expressions can be rewritten in different forms using the properties of operations.

The following properties of operations apply to all rational numbers.

The ***distributive property of multiplication over addition or subtraction*** states that a factor is "distributed" to all terms within a set of parentheses.

The ***commutative property*** states that for addition and multiplication, the order in which the operation is performed does not change the value of the expression.

The ***associative property*** states that for addition and multiplication, the way in which numbers are grouped does not change the value of the expression.

9.6.2 GUIDED INSTRUCTION: REWRITING EQUIVALENT EXPRESSIONS

TEACHER MOVES

Consider having students participate in a Notice and Wonder routine to give formal space for processing that leads to accuracy in evaluating. Students want to "just find the answer," which often leads to errors.

ENRICHMENT

Discuss whether it is necessary to evaluate each expression to determine if the expressions are equivalent.

- Do we need to evaluate the expression to determine equivalence?
- How would you know they are equivalent without evaluating?

TEACHER MOVES

Consider having students engage in a Turn and Talk after each definition to rephrase the definition of each property in their own words and provide an example illustrating their definition.

DIFFERENTIATE INSTRUCTION

Consider a foldable, Frayer Model, or type of interactive notebook for this vocabulary. Examples, nonexamples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

9.6.2 GUIDED INSTRUCTION: REWRITING EQUIVALENT EXPRESSIONS (CONTINUED)

QUESTIONING

Challenge students who finish early to rewrite the expression in three different ways using all three properties.

6. Work with your partner to rewrite the expression $5 - 6.2$ using the commutative property.

   $-6.2 + 5$

7. Work with your partner to rewrite the following expression in **two** equivalent forms using the properties of operations.

$$\left(\frac{2}{5} + 4\right) + 6 - \frac{1}{2}(5 - 6.2)$$

Answers will vary. Two examples include:

$= \left(4 + \frac{2}{5}\right) + 6 - \frac{1}{2}(5 - 6.2)$

$= \left(\frac{2}{5} + 4\right) + 6 - 2.5 + 3.1$

9.6.3 Try It: Rewriting Equivalent Expressions

1. Maria was asked to rewrite the following expression in a variety of ways.

$$5\frac{1}{4} - 4(-3.5 + 7) + \left(\frac{3}{5} + 5\right) + 8.25$$

She came up with four different expressions, but she made some errors. Without evaluating, decide which expressions are equivalent using properties of operations. Then, identify the errors she made.

Expression	Is it equivalent? Justify why or why not.
$5\frac{1}{4} - 14 - 28 + \left(\frac{3}{5} + 5\right) + 8.25$	*No, because the sign of the 14 should be positive. When using the distributive property to multiply -4 by -3.5, we obtain 14 not -14.*
$5\frac{1}{4} - 4(-3.5 + 7) + \frac{3}{5} + (8.25 + 5)$	*Yes, because of the commutative and associative properties of addition.*
$5\frac{1}{4} + 14 - 28 + \frac{3}{5} + (5 + 8.25)$	*Yes, because of the distributive property of multiplication over addition and the associative property.*
$1.25(-3.5 + 7) + \left(5 + \frac{3}{5}\right) + 8.25$	*No, because the order of operations was not followed. The student subtracted 4 from $5\frac{1}{4}$ before multiplying -4 by $(-3.5 + 7)$.*

9.6.3 TRY IT: REWRITING EQUIVALENT EXPRESSIONS

ENRICHMENT

Refer students to 9.6.2 Guided Instruction question 1 as scaffolding for determining equivalence without evaluating.

TEACHER MOVES

Monitor for responses like “because they are the same.” or “because they are not the same.” Use talk moves to extend student thinking: “Say more. What exactly is not the same? What does that have to do with equivalence?”

DIFFERENTIATE INSTRUCTION

Consider providing mark-up tools like highlighters, colored pencils, or markers for students to visually represent their findings.

9.6.4 YOUR TURN!

TEACHER MOVES

Consider having students pair with a partner to check each other's work once their work is completed. Encourage students to determine which equivalent expressions represent each property and make sure they have represented all three properties between them. If not, they can work together to represent any missing properties.

QUESTIONING

Challenge advanced learners to create expressions that use both decimals and fractions and all three properties.

9.6.4 Your Turn!

1. Rewrite the following expression in **two** ways using the properties of operations.

$$\frac{12}{35} \cdot \frac{1}{2} + \frac{1}{5} + \left(\frac{2}{5} + \frac{3}{7}\right)$$

Answers will vary. Examples include:

$= \frac{12}{35} \cdot \frac{1}{2} + \frac{1}{5} + \left(\frac{3}{7} + \frac{2}{5}\right)$ *Commutative property of addition*

$= \frac{12}{35} \cdot \frac{1}{2} + \left(\frac{1}{5} + \frac{2}{5}\right) + \frac{3}{7}$ *Associative property of addition*

$= \frac{1}{2} \cdot \frac{12}{35} + \frac{1}{5} + \left(\frac{2}{5} + \frac{3}{7}\right)$ *Commutative property of multiplication*

9.6.5 Wrap-Up: Error Analysis

1. Juan was asked to rewrite the expression $\left(-6 + 2\frac{1}{2} + 3\right) + 4 - 2\left(-3\frac{2}{3}\right)$ using the properties of operations.

   Juan wrote $-6 + \left(3 + 2\frac{1}{2} + 4\right) + 7\frac{1}{3}$.

   Determine if Juan is correct. Show your work or explain your thinking.

   Yes, because he used the commutative and associative properties of addition and evaluated the multiplication before the addition (using the order of operations).

9.6.5 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

Notice the prompt asks students to show their work OR explain their thinking, providing different entry points for a variety of students. Encourage students who show their work to highlight, circle, or mark-up their work to show the errors.